

CARLOS LORENZO-ZÚÑIGA MARÍ

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EDUCATION

- **Universitat Politècnica de València** Valencia, ES
Degree in Biomedical Engineering *Sep 2025 – Jun 2029*
 - **Focus Areas:** Embedded Systems, Biosignal Processing, Neurotechnology.

EXPERIENCE

- **Daidalonic** Valencia, ES
Signal Processing Engineer *Oct 2025 – Present*
 - **Hardware Engineering:** Engineered 2 multi-layer sEMG data acquisition PCBs incorporating galvanic isolation and ESD protection, enabling ~99% muscle contraction detection accuracy in downstream edge ML models (Random Forest/CNNs).
 - **Analog Front-End Design:** Designed 2-stage active analog front-ends featuring 30-300Hz bandpass filtering and DRL bias, achieving up to 1100V/V amplification and robust RMS/offset signal conditioning.
 - **Hardware Validation:** Standardized hardware validation by authoring DVT protocols, expanding testing bandwidth across the team and successfully isolating and resolving a critical power supply reliability issue.
- **Caxton College** Puzol, ES
IT Specialist *Jun 2025 – Jul 2025*
 - **Infrastructure Automation:** Automated a domain controller (Active Directory) migration for 1,700 students, reducing a 3-day multi-technician process to 30 minutes and saving thousands of euros in labor costs.
 - **Custom Tooling:** Developed a custom Google Apps Script platform for faculty to report and monitor over 1,000 students, streamlining communication workflows via database fuzzy-finding and automated emailing.

PROJECTS

- **8-Channel EEG Headset (Bespoke BCI)**
Embedded Systems & Edge ML Project
 - **Hardware Architecture:** Designed a custom 4-layer PCB integrating a TI ADS1299 AFE and ESP32-S3, implementing active DRL bias circuitry and digital filtering (50Hz notch, 4-38Hz bandpass).
 - **Cost Optimization:** Reduced prototype BOM to 100 EUR (250 EUR total turn-key PCBA), delivering an advanced neurotech platform at <10% the cost of equivalent commercial alternatives.
 - **Edge Deployment:** Authored a bespoke ESP-IDF driver for the ADS1299; optimized and quantized an EEGNet architecture for 8-bit inference on-chip, delivering 65% accuracy on 4-class motor imagery datasets.
- **VECTRA: Custom C++ Physics Engine**
Graphics & Systems Programming
 - **Architecture:** Programmed a rigid-body physics engine from scratch in C++ utilizing an Entity Component System (ECS) architecture and clean object-oriented design.
 - **Collision Pipeline:** Implemented a discrete impulse-based solver for manifold generation, resolving complex multi-body contact constraints smoothly at a locked 300Hz simulation tick rate.

TECHNICAL SKILLS

- **Hardware & Embedded:** PCB Design (KiCad, EasyEDA), ESP-IDF Firmware, Microcontrollers (ESP32-S3), ADS1299, Soldering, Oscilloscopes, Function Generators, DVT protocols.
- **Signal Processing:** sEMG/EEG Acquisition, Analog/Digital Filtering, Amplification, DRL
- **Software & Languages:** C/C++, Python (PyTorch, NumPy, Pandas), C#, TypeScript, Git, Linux.
- **Languages:** Spanish (Native), English (C2 Cambridge), German (B2 Goethe).